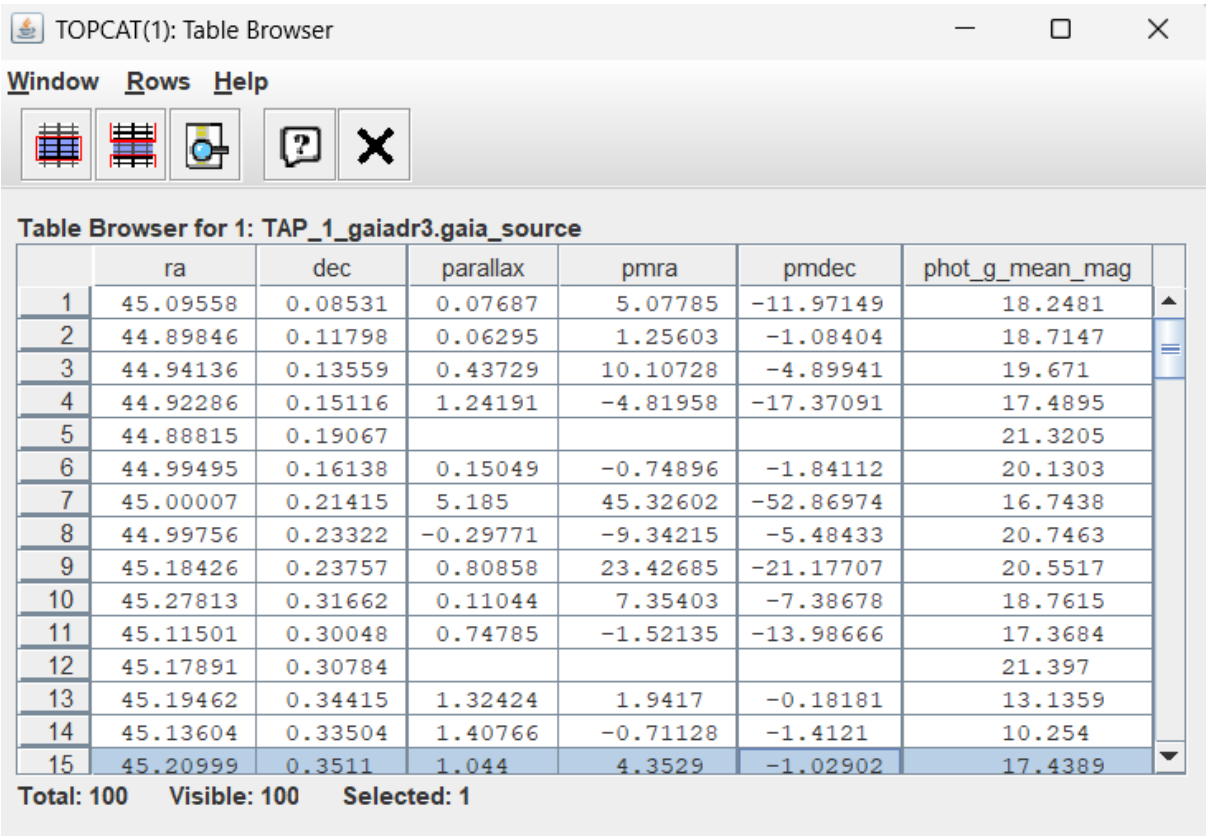
TOPCAT – Working with the Software

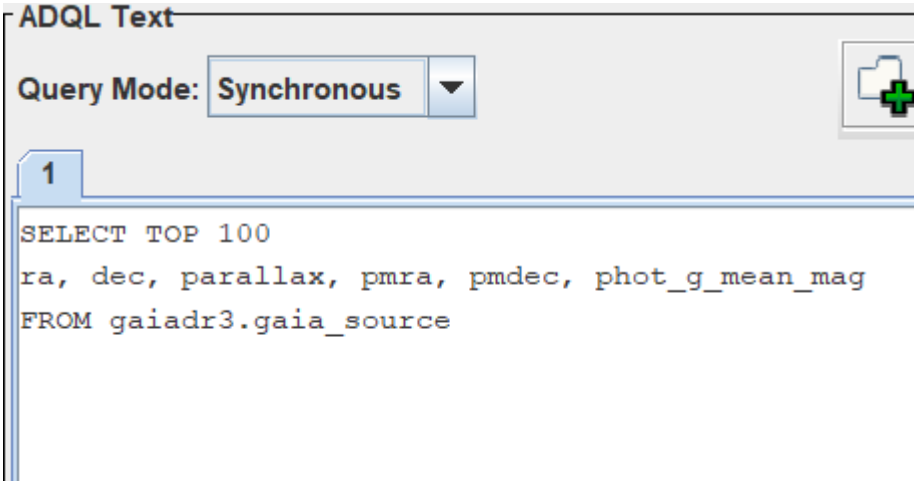
Stellar Astronomy

Extracting Data from a data service

1. On the TOPCAT Software we invoke the TAP Query.
2. It opens a window with the list of all available services which we can then select.
3. I selected the Gaia data, lite version.
4. Now select a table from the Gaiadr3 data folder in the Gaia service module.
5. Using SQL you can extract relevant stars that you want.

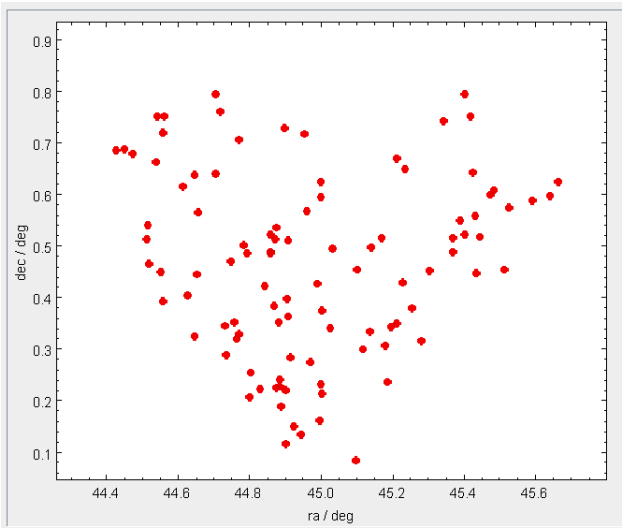


	ra	dec	parallax	pmra	pmdec	phot_g_mean_mag
1	45.09558	0.08531	0.07687	5.07785	-11.97149	18.2481
2	44.89846	0.11798	0.06295	1.25603	-1.08404	18.7147
3	44.94136	0.13559	0.43729	10.10728	-4.89941	19.671
4	44.92286	0.15116	1.24191	-4.81958	-17.37091	17.4895
5	44.88815	0.19067				21.3205
6	44.99495	0.16138	0.15049	-0.74896	-1.84112	20.1303
7	45.00007	0.21415	5.185	45.32602	-52.86974	16.7438
8	44.99756	0.23322	-0.29771	-9.34215	-5.48433	20.7463
9	45.18426	0.23757	0.80858	23.42685	-21.17707	20.5517
10	45.27813	0.31662	0.11044	7.35403	-7.38678	18.7615
11	45.11501	0.30048	0.74785	-1.52135	-13.98666	17.3684
12	45.17891	0.30784				21.397
13	45.19462	0.34415	1.32424	1.9417	-0.18181	13.1359
14	45.13604	0.33504	1.40766	-0.71128	-1.4121	10.254
15	45.20999	0.3511	1.044	4.3529	-1.02902	17.4389

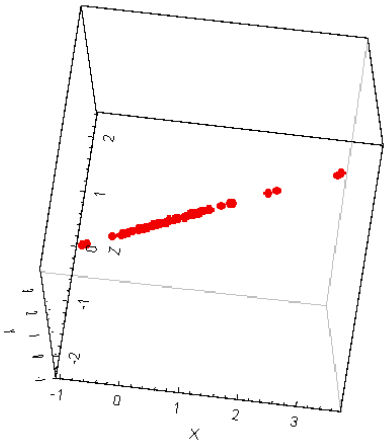


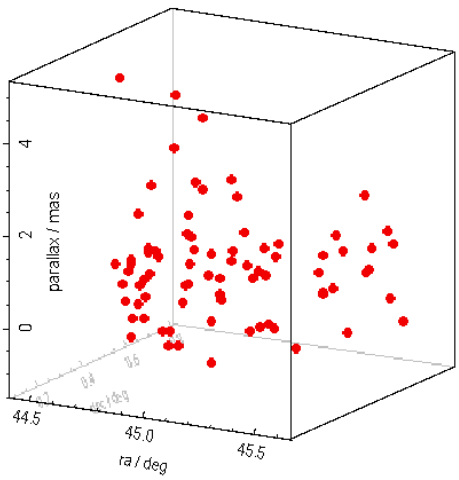
6. This opens up a separate window where you can see the table also

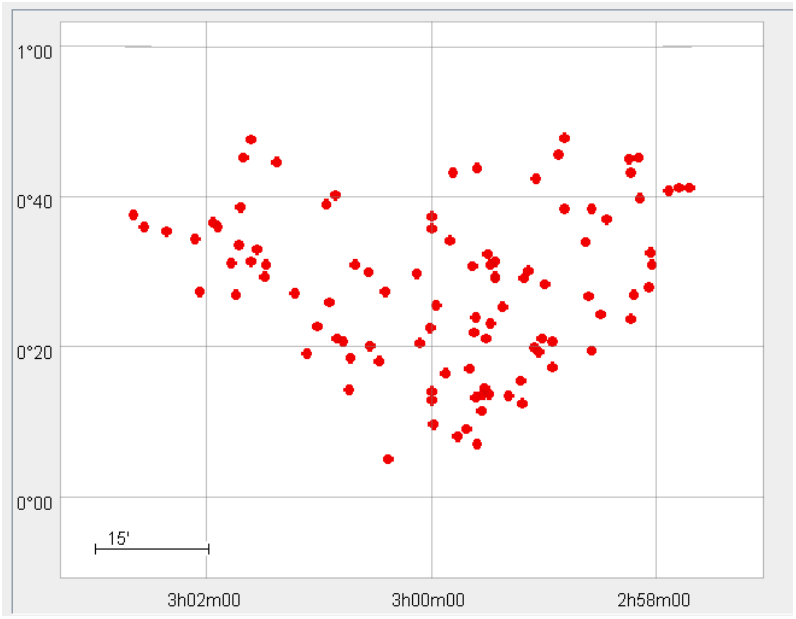
Creating Plots



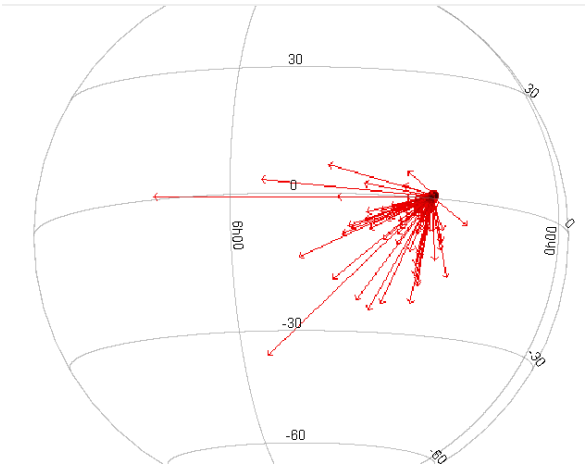
7. You can also plot the stars in different dimensions, 2D or 3D plots. Cartesian or Spherical.

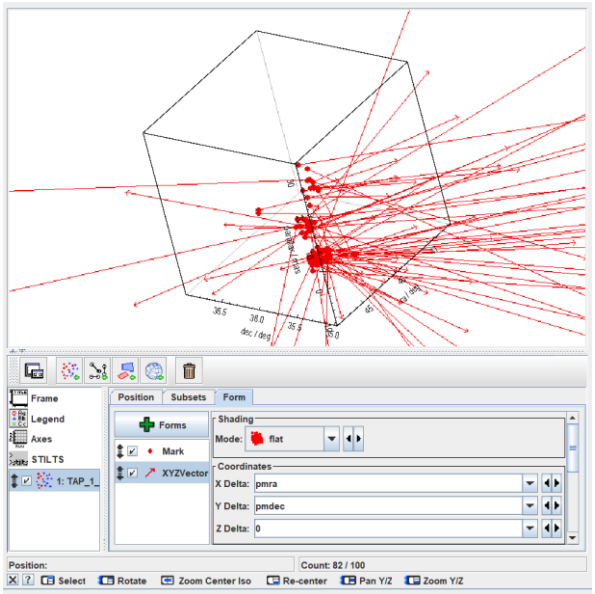






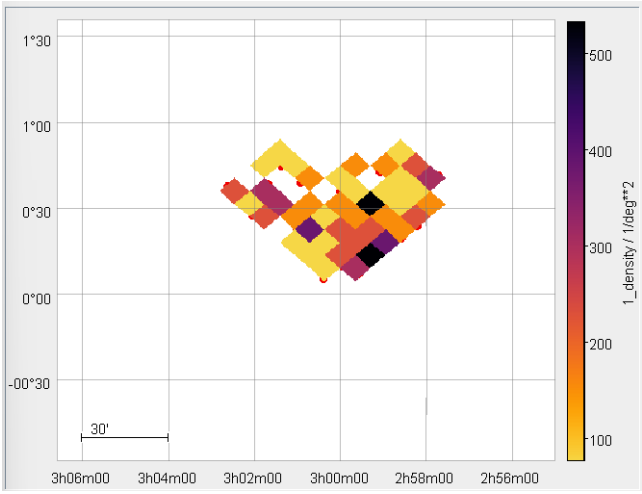
8. We can plot the proper motions of the stars. The arrows with varying sizes show the magnitude of the velocities of these stars.





a. Some stars are moving with extremely high velocities compared to the other stars. **High Velocity Stars.**
b. Some are moving in same direction with the same speed and are in close proximity. That is a **Star Cluster**.

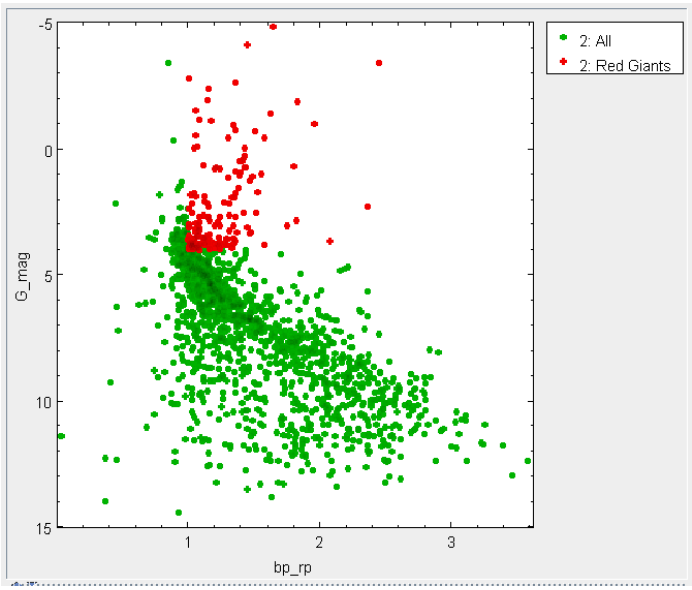
9. Sky density: the darker the region, more the number of star density.



10. Subsets can be created in the TOPCAT main window by defining an expression and then subsets in the plot can be distinguished by changing the colour of the dots.

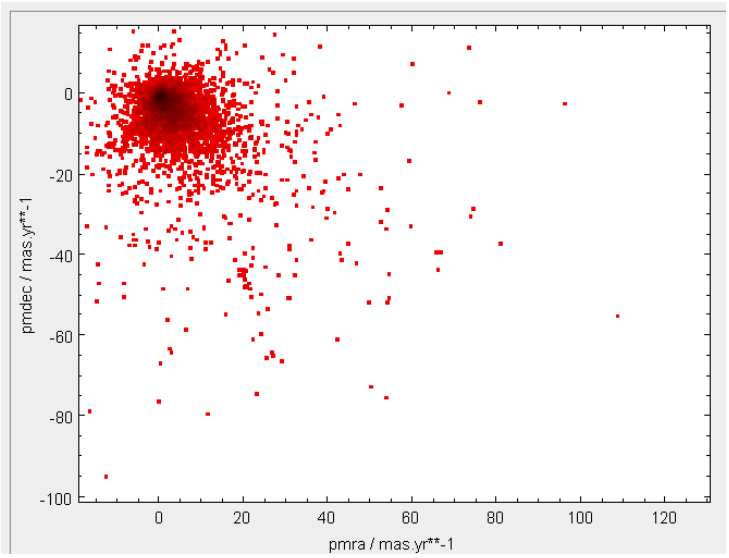
Creating HR Diagrams

11. We can also create an HR Diagram. We first extract the apparent magnitudes, the blue and red colour magnitudes. We then plot a graph between apparent magnitudes and colour. Then create a subsets of Red giants and white dwarfs and etc.

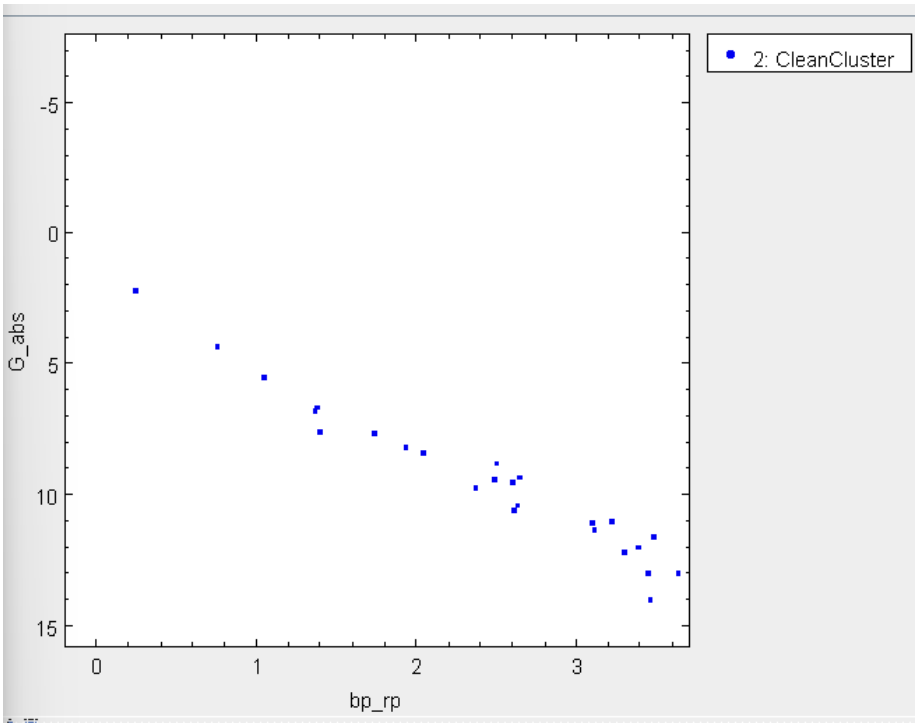


12. The HR Diagram is extremely messy so we target star clusters like the M45 and erase the rest. We scan for stars in the region which has the plaeides cluster. Then we remove the other useless stars.

13. Plot the graph for pmdec and pmra and notice a knot at around 20,-45. This is plaeides.



14. We extract this cluster: Plaedies cluster. Create a subset of this and then plot the HR plot of this. We further get a main sequence of the Plaedeis Cluster.

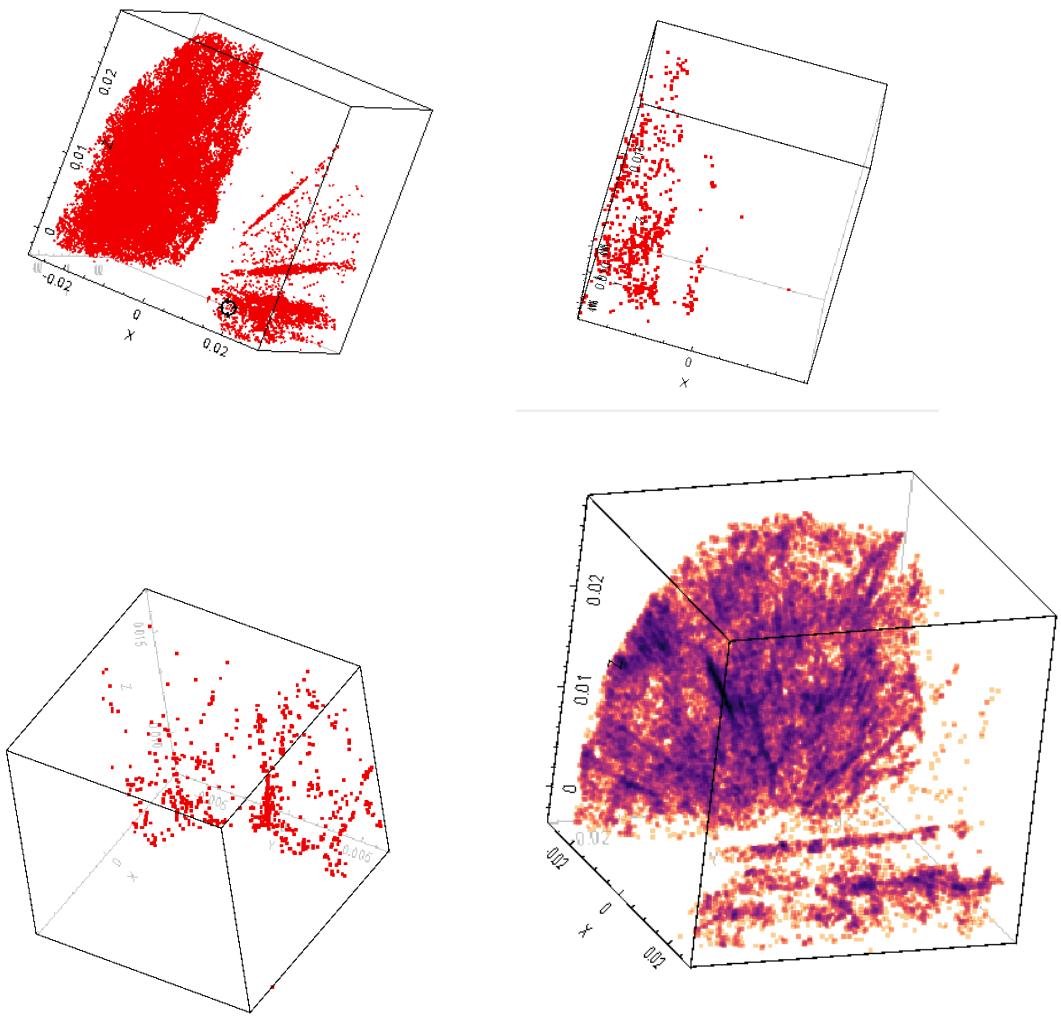


Galactic Web

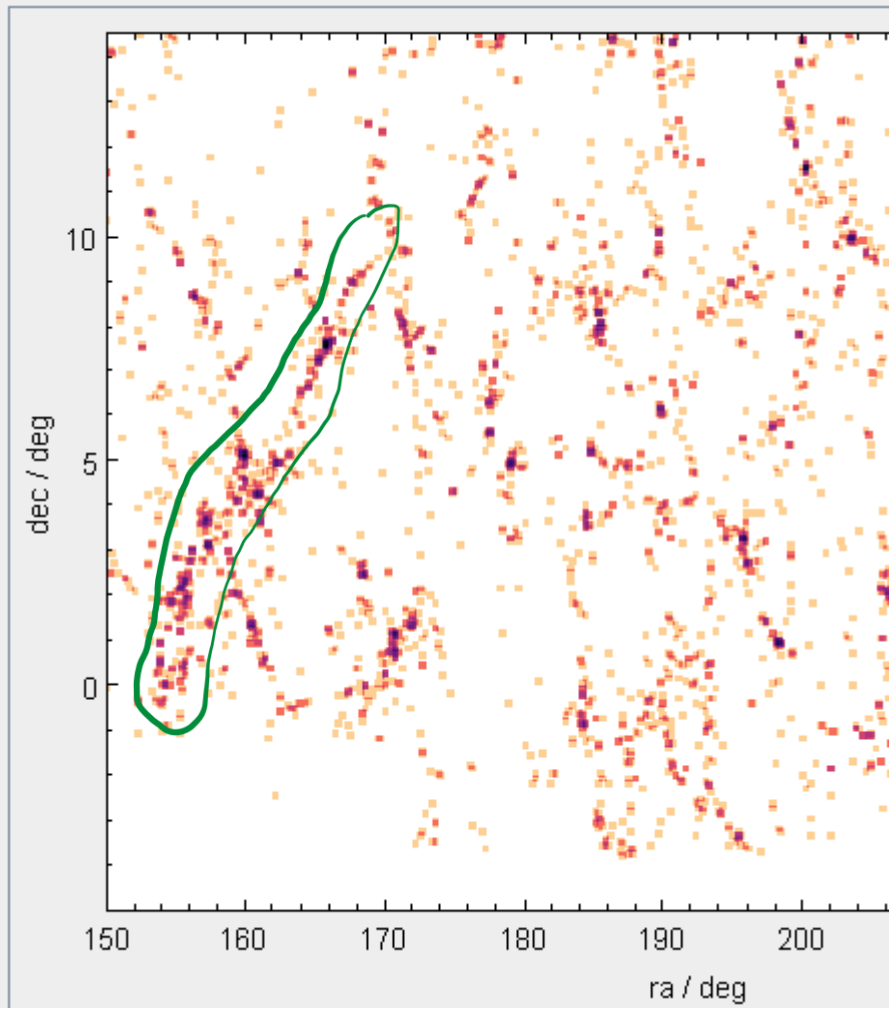
Initially we were trying to work with Millennium Run data that has simulations of the galactic webs. But the data wasn't available as the servers were down. So as an alternative I tried to download raw data from another server on the TAP Query Window from the SDSS Data Release Table. We extract 50,000 galaxy data and using a 3D cube plot build the "SDSS cone plot."

It contains the North Galactic Cap and the South Galactic Cap. One can see if they zoom into the wedges, galaxies arranged in the form of filaments and voids in between the galaxy filaments.

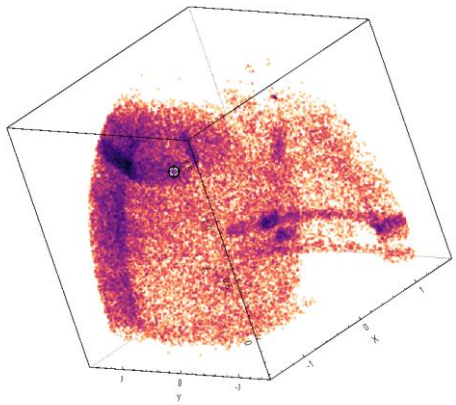
One can see large gaps between the fans and this is because of our Milky way galaxy. It creates a ZONE OF AVOIDANCE. IF we ignore the Milky way then it would have probably been a spherical arrangement of Galaxies with filaments and voids. This is because of **The Cosmological Principle**.



One of the largest known structures previously was the Sloan Great Wall. It is present at a redshift of >0.07 to <0.085 from us. I tried to capture the wall in isolation.



If we increase the redshift we can go back into time when the universe was still forming, before the formation of the Cosmic Web. We can see more uniformity and less formation of clumps.



Simple Image Access Query

We can connect the SAMP Hub to another terminal such as ds9 to be able the image in that while extracting FIT files from TOPCAT.

OR

Manually export the fit files into ds9 through URL links.

Utilised the above methods to see M51 galaxy through the DSS2 Data set. Or the Crab Nebula.

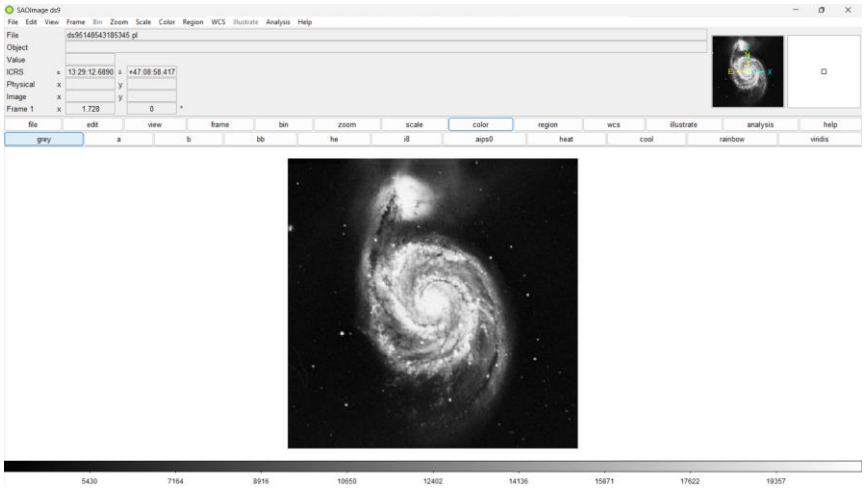
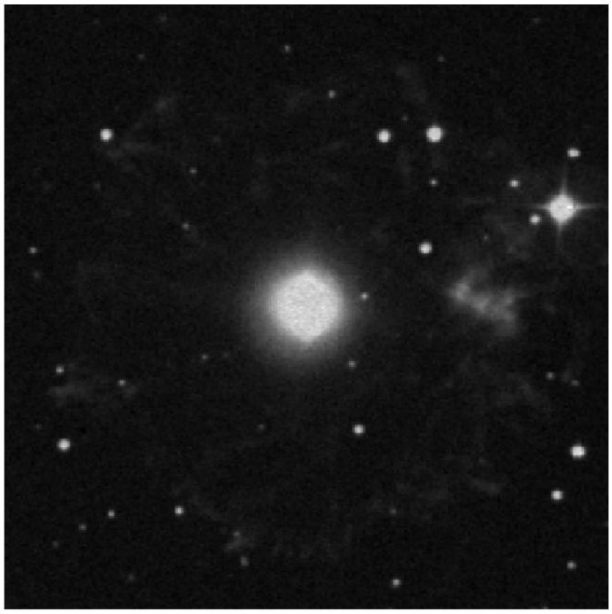


Table Browser for 4: M51-DSS2-12m

	Survey	Ra	Dec	Dim	Size	Scale	Format	PixFlags	URL	LogicalName
1	dss2b	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/fits	F	tes=J2000.0&requestID=skv1767005365074&return=FITS	1
2	dss2b	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/tiff	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	1
3	dss2b	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/gif	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	1
4	dss2b	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/png	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	1
5	dss2b	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/jpeg	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	1
6	dss2r	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/fits	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	2
7	dss2r	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/tiff	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	2
8	dss2r	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/gif	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	2
9	dss2r	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/png	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	2
10	dss2r	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/jpeg	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	2
11	dss2ir	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/fits	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	3
12	dss2ir	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/tiff	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	3
13	dss2ir	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/gif	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	3
14	dss2ir	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/png	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	3
15	dss2ir	202.46957	47.19526	2	(300, 300)	(-6.6666666666666668E-4, 6.6666666666666668E-4)	image/jpeg	F	https://skyview.gsfc.nasa.gov/cgi-bin/images?position=202	3

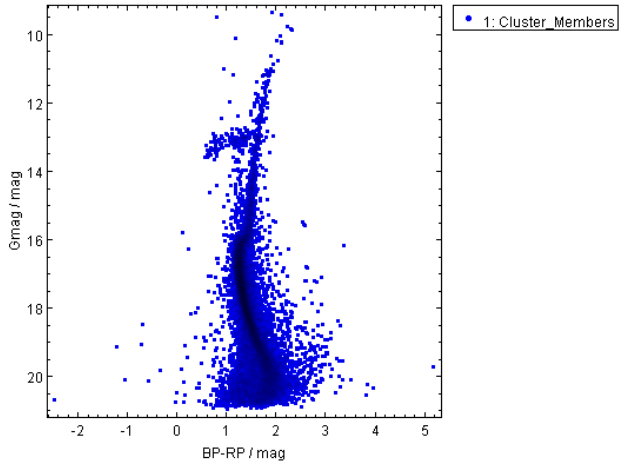


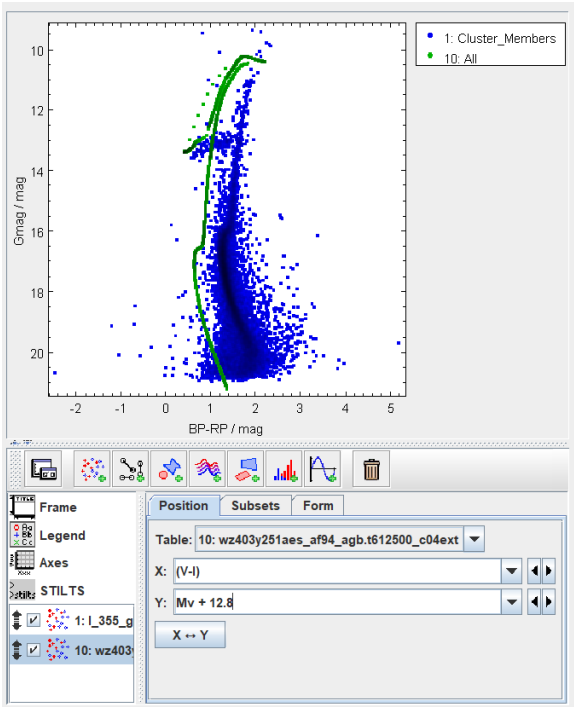
To check if Isochrones Fit

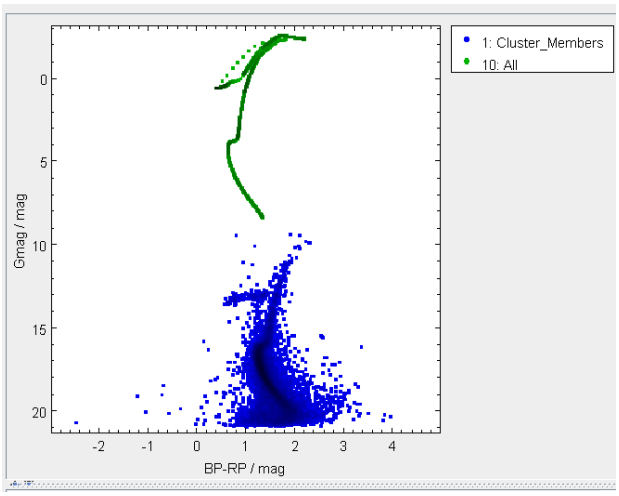
BaSTi is a high-level physics experiment which is a collection of theoretical datasets. It is shown in green colour in the graph. Here I was trying to compare the theoretical and experimental plot of the CMD for the M4 Globular Cluster. The experimental data was acquired from the Vizier Service Catalogue from the Gaia DR3. The BaSTi Data Loader provided us with Johnson Castelli Data. To get the M4 Globular Cluster data we set the age to 12.5 Gya, Metallicity of roughly -1.05, Mass load of 0.4, etc.

The plot of Vizier was generated and on top of it layering of another plot was done and both were compared.

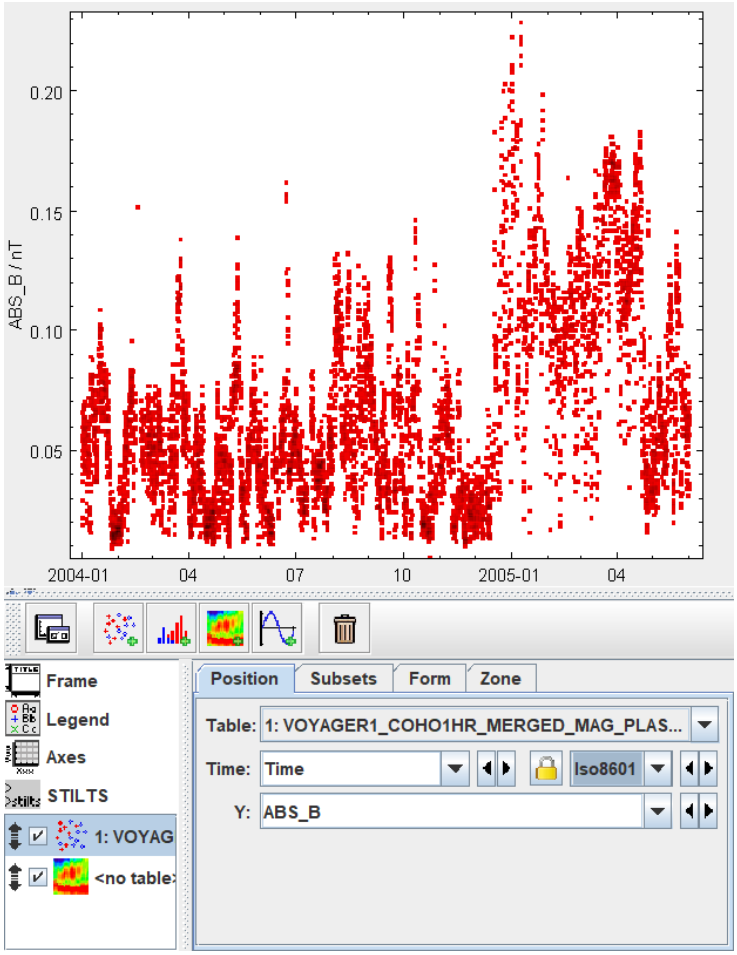
The pattern of the data from both the plots look similar.







HAPI Query Window

Used for working with Time series Data (non-static data) such as Plasma, Magnetic flux, Solar Winds. More prevalent in Solar Astronomy and Helio physics.

We worked on Solar winds data acquired by the Voyager 1 Space craft when it was crossing the Termination Shock i.e. escaping the Solar System into Interstellar space.